son.pu.py - C:/Users/pings/Desktop/son.pu.py (3.14.2)

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```
#Z-Test
#1. For the population at large, the Wechsler Adult Intelligence Scale is designed to yield a normal distribution of test scores
#with a mean of 100 and a standard deviation of 15.
#School district officials wonder whether, on the average, an IQ score different from 100 describes the intellectual aptitudes
#of all students in their district. Wechsler IQ scores are obtained for a random sample of 25 of their students,
#and the mean IQ is found to equal 105. Using the step-by-step procedure, test the null hypothesis at the .05 level of significance. (Two-tailed)
#Table value: Z (two-tailed, α = 0.05) = ±1.96

import math

mu0 = 100
xbar = 105
sigma = 15
n = 25
alpha = 0.05
table = 1.96

z = (xbar - mu0) / (sigma / math.sqrt(n))

print("Problem 1: IQ of students (Z-test, two-tailed)")
print("H0: mu = 100")
print("H1: mu != 100")
print("Level of significance:", alpha)
print("Test statistic: Z =", round(z, 4))
print("Table value: +-", table)

if abs(z) > table:
    print("Decision: Reject H0")
    print("Conclusion: The mean IQ is significantly different from 100.")
else:
    print("Decision: Fail to reject H0")
    print("Conclusion: The mean IQ is not significantly different from 100.")
```

Ln: 32 Col: 77

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/pings/Desktop/son.pu.py ==========
Problem 1: IQ of students (Z-test, two-tailed)
H0: mu = 100
H1: mu != 100
Level of significance: 0.05
Test statistic: Z = 1.6667
Table value: +- 1.96
Decision: Fail to reject H0
Conclusion: The mean IQ is not significantly different from 100.
>>>
```

Ln: 13 Col: 0